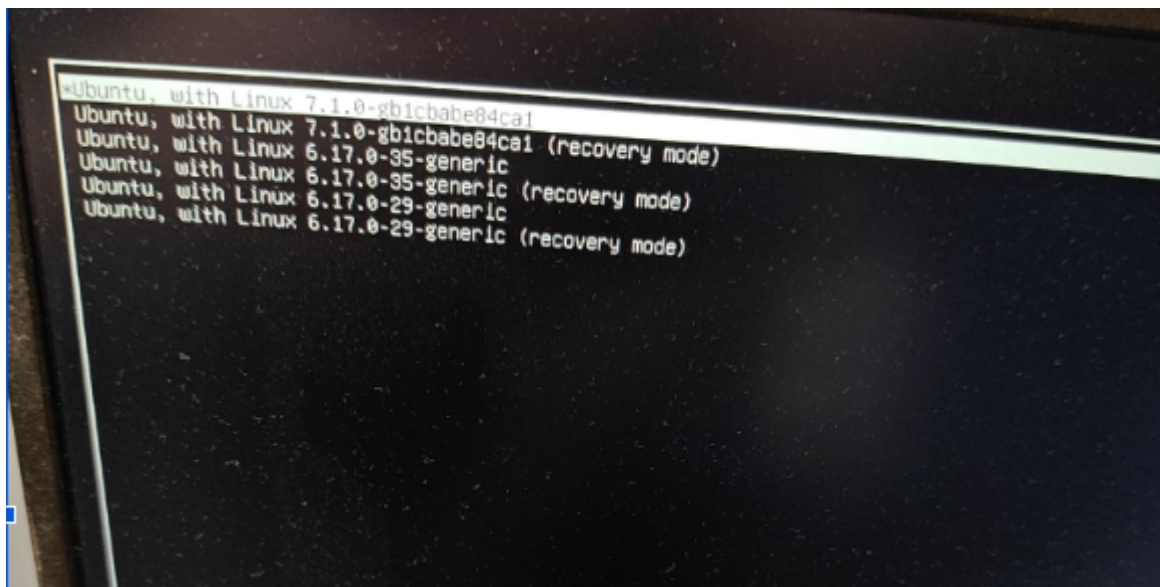


- 1) Собрал дефолтный конфиг для моего железа
make defconfig
- 2) Собрал ядро и модули
make -j6 bzImage && make -j6 modules
- 3) Установил ядро и модули
sudo make modules_install
sudo make install

текущее ядро: grogu@osio:/boot\$ uname -r
6.17.0-35-generic

новое ядро: initrd.img-7.1.0



grub увидел новое ядро.

- 1) Устанавливаем кросс компилятор для arm
sudo apt install gcc-arm-linux-gnueabi

2) Собираем дефолтный конфиг для arm

ARCH=arm make defconfig

3) Собираем ядро

make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- -j6 zImage

4) ядро собрано

```
grogu@osio: ~/project/kernel/linux
```

```
LD      vmlinux.unstripped
NM      System.map
SORTTAB vmlinux.unstripped
OBJCOPY vmlinux
GEN     modules.builtin.modinfo
GEN     modules.builtin
OBJCOPY arch/arm/boot/Image
Kernel: arch/arm/boot/Image is ready
LDS     arch/arm/boot/compressed/vmlinux.lds
AS      arch/arm/boot/compressed/head.o
GZIP    arch/arm/boot/compressed/piggy_data
CC      arch/arm/boot/compressed/misc.o
CC      arch/arm/boot/compressed/decompress.o
CC      arch/arm/boot/compressed/string.o
AS      arch/arm/boot/compressed/hyp-stub.o
CC      arch/arm/boot/compressed/fdt_rw.o
CC      arch/arm/boot/compressed/fdt_ro.o
CC      arch/arm/boot/compressed/fdt_wip.o
CC      arch/arm/boot/compressed/fdt.o
CC      arch/arm/boot/compressed/atags_to_fdt.o
CC      arch/arm/boot/compressed/fdt_check_mem_start.o
AS      arch/arm/boot/compressed/lib1funcs.o
AS      arch/arm/boot/compressed/ashldi3.o
AS      arch/arm/boot/compressed/bswapsdi2.o
AS      arch/arm/boot/compressed/piggy.o
LD      arch/arm/boot/compressed/vmlinux
OBJCOPY arch/arm/boot/zImage
Kernel: arch/arm/boot/zImage is ready
grogu@osio:~/project/kernel/linux$
```

5) Собираем dtbs файлы

ARCH=arm make -j6 dtbs

```

DTC arch/arm/boot/dts/nxp/imx/imx7s-colibri-aster.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7s-colibri-eval-v3.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7s-colibri-iris.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7s-colibri-iris-v2.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7s-mba7.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7s-warp.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7ulp-com.dtb
DTC arch/arm/boot/dts/nxp/imx/imx7ulp-evk.dtb
OVL arch/arm/boot/dts/nxp/imx/imx53-qsb-hdmi.dtb
OVL arch/arm/boot/dts/nxp/imx/imx53-qsr-b-hdmi.dtb
OVL arch/arm/boot/dts/nxp/imx/imx6qdl-dhcom-pdk2-overlay-497-200-x12.dtb
OVL arch/arm/boot/dts/nxp/imx/imx6qdl-dhcom-pdk2-overlay-505-200-x12-ch1010hlwh.dtb
OVL arch/arm/boot/dts/nxp/imx/imx6qdl-dhcom-pdk2-overlay-531-100-x21.dtb
OVL arch/arm/boot/dts/nxp/imx/imx6qdl-dhcom-pdk2-overlay-531-100-x22.dtb
OVL arch/arm/boot/dts/nxp/imx/imx6qdl-dhcom-pdk2-overlay-560-200-x12.dtb
OVL arch/arm/boot/dts/nxp/imx/imx6qdl-dhcom-picoitx-overlay-626-100-x2-ch1010hlwh.dtb
rogu@osio:~/project/kernel/linux$

```

6) Запускаем qemu без загрузчика ОС

QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic

```

grogu@osio: ~/project/test
5.637118] List of all bdev filesystems:
5.637147] ext3
5.637169] ext2
5.637180] ext4
5.637192] squashfs
5.637204] vfat
5.637215] msdos
5.637227] ntfs
5.637244]
5.637602] Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0)
5.648575] CPU: 0 UID: 0 PID: 1 Comm: swapper/0 Tainted: G      W      7.1.0-07602-g66affa37cfac #1 VOLL
5.649348] Tainted: [W]=WARN
5.649580] Hardware name: ARM-Versatile Express
5.649939] Call trace:
5.650095] unwind_backtrace from show_stack+0x10/0x14
5.650702] show_stack from dump_stack_lvl+0x54/0x68
5.651085] dump_stack_lvl from vpanic+0x234/0x450
5.651432] vpanic from do_panic_on_target_cpu+0x0/0x14
5.653104] ---[ end Kernel panic - not syncing: VFS: Unable to mount root fs on unknown-block(0,0) ]---

```

Ядро упало, так как нет бинарника init. И за за того что нет корневой файловой системы.